Sixteen-year-old Tabinshweti succeeded his father as ruler in 1531. Not content with ruling Upper Burma, Tabinshweti planned on expanding Toungoo by invading Pegu. Controlling Pegu would give Toungoo access to its abundant natural wealth and direct access to the international maritime trade and foreign mercenaries. He captured the Mon port city of Bassein in the Ayeyarwaddy delta in 1535. Pegu finally fell in 1539 when Tabinshweti occupied the city and forced the Mon king Tushintakayutpi (r. ca. 1526–1538) to flee to Prome. He then transferred the capital from Toungoo to Pegu and was recognized as the undisputed king of Lower Burma. Tabinshweti then held a double coronation ceremony in both Pegu and Bagan to demonstrate that he ruled both Upper and Lower Burma. He built his palace on the site of the old Mon city. Following the unification of Burma in 1539, Tabinshweti attempted two unsuccessful campaigns against Tavoy and Ayutthaya, the Siamese capital in 1548, only to be forced to return to Pegu. Humiliated by the crushing defeats Tabinshweti retreated from public life before being assassinated by a close Mon advisor Smim Sawhtut in 1551. Smim Sawhtut crowned himself king but was in turn killed by Smim Htaw, the son of the last king of Pegu, who led a Mon army to Pegu, where Smim Sawhtut was executed after ruling for only three months.

Tabinshweti's brother-in-law and eventual successor King Bayinnaung (r. ca. 1551–1581) crushed the Mon revolt in Toungoo, killed Smim Htaw, and installed himself as king. He then led his army south, capturing the city of Pegu and executing Smim Htaw, the son of the last Mon king of Pegu. Under Bayinnaung's rule, the port of Pegu would come to dominate the maritime trade of mainland Southeast Asia. Following the consolidation of control of both Upper Burma and Lower Burma, Bayinnaung set about expanding his empire. Bayinnaung launched a series of military campaigns from Upper Burma that culminated in Toungoo's control of Ayutthaya, Lon Chang, and the Shan states. In 1555, he defeated the Shan at Ava, placing them under Burmese suzerainty for the first time. Between 1555 and 1557, Bayinnaung led two additional campaigns in which he conquered the Shan states to the north and east. Alarmed by Bayinnaung's victory over the Shans, the Tai of Chiang Mai formulated rebellion among the Shan Sawbwas. Bayinnaung's response to the Tai interference was swift. He crossed the Salween River and conquered Chiang Mai in 1558.